

### Mehanske karakteristike:

Beton: C40/50

$$f_{ck} = 4,0 \text{ kN/cm}^2$$

$$\gamma_c = 1,5$$

$$f_{cd} = 2,67 \text{ kN/cm}^2$$

$$E_{cm} = 3500 \text{ kN/cm}^2$$

$$f_{ctm} = 0,35 \text{ kN/cm}^2$$

Armatura: B500B

$$f_{yk} = 50 \text{ kN/cm}^2$$

$$\gamma_s = 1,15$$

$$f_{yd} = 43,48 \text{ kN/cm}^2$$

$$E_s = 20000 \text{ kN/cm}^2$$

### Geometrijske karakteristike:

$$b = 30 \text{ cm}$$

$$h = 60 \text{ cm}$$

$$\text{Krovni sloj: } 30 \text{ mm}$$

$$d_g = 16 \text{ mm}$$

$$a = 4,8 \text{ cm}$$

$$d = 55,2 \text{ cm}$$

### Obremenitev:

$$M_{Ed} = 258,07 \text{ kNm}$$

$$N_{Ed} = 0 \text{ kN}$$

$$z_s = 25 \text{ cm}$$

$$M_{Eds} = 258,1 \text{ kNm}$$

### Potrebna vzdolžna armatura:

$$k_d = 0,106$$

$$k_{s,Rogač} = 1,068$$

$$k_{s,EC2} = 1,061$$

$$A_{s,potr.} = 11,48 \text{ cm}^2$$

### Minimalna vzdolžna armatura:

$$A_{s,min} = 3,01 \text{ cm}^2$$

### Minimalna razdalja med sosednjima palicar

$$e_{min} = 21 \text{ mm}$$

### Maksimalno število palic v eni vrsti:

$$n_{max} = 6,51$$

### Izberemo:

$$n = 6$$

$$\phi_{vzd} = 16 \text{ mm}$$

$$\phi_{streme} = 10 \text{ mm}$$

$$A_{s,dej.} = 12,06 \text{ cm}^2$$

$$e = 4,4 \text{ cm}$$

Pravokotni prerez - enojna armatura

Tabela 2

| $\varepsilon_s$ [%] |       | $-\varepsilon_1$ [%] |       |       |       |       |       |       |       |       |       |       |
|---------------------|-------|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                     |       | 1.00                 | 1.25  | 1.50  | 1.75  | 2.00  | 2.25  | 2.50  | 2.75  | 3.00  | 3.25  | 3.50  |
| 1.5                 | $k_x$ | .400                 | .455  | .500  | .538  | .571  | .600  | .625  | .647  | .667  | .684  | .700  |
|                     | $k_s$ | 1.163                | 1.193 | 1.220 | 1.247 | 1.273 | 1.298 | 1.323 | 1.347 | 1.370 | 1.391 | 1.411 |
|                     | $k_h$ | .143                 | .189  | .230  | .268  | .299  | .325  | .346  | .364  | .379  | .391  | .402  |
| 3.0                 | $k_x$ | .250                 | .294  | .333  | .368  | .400  | .429  | .455  | .478  | .500  | .520  | .538  |
|                     | $k_s$ | 1.096                | 1.117 | 1.137 | 1.157 | 1.176 | 1.196 | 1.216 | 1.235 | 1.254 | 1.272 | 1.289 |
|                     | $k_h$ | .095                 | .130  | .165  | .197  | .227  | .252  | .274  | .293  | .310  | .325  | .338  |
| 4.0                 | $k_x$ | .200                 | .238  | .273  | .304  | .333  | .360  | .385  | .407  | .429  | .448  | .467  |
|                     | $k_s$ | 1.075                | 1.092 | 1.109 | 1.126 | 1.143 | 1.160 | 1.177 | 1.194 | 1.210 | 1.226 | 1.241 |
|                     | $k_h$ | .078                 | .108  | .138  | .168  | .194  | .218  | .240  | .259  | .276  | .291  | .304  |
| 5.0                 | $k_x$ | .167                 | .200  | .231  | .259  | .286  | .310  | .333  | .355  | .375  | .394  | .412  |
|                     | $k_s$ | 1.062                | 1.076 | 1.091 | 1.105 | 1.120 | 1.135 | 1.150 | 1.165 | 1.179 | 1.193 | 1.207 |
|                     | $k_h$ | .065                 | .092  | .119  | .145  | .170  | .192  | .213  | .231  | .247  | .262  | .276  |
| 6.0                 | $k_x$ | .143                 | .172  | .200  | .226  | .250  | .273  | .294  | .314  | .333  | .351  | .368  |
|                     | $k_s$ | 1.053                | 1.065 | 1.078 | 1.091 | 1.103 | 1.117 | 1.130 | 1.143 | 1.156 | 1.169 | 1.181 |
|                     | $k_h$ | .057                 | .080  | .104  | .128  | .151  | .172  | .191  | .208  | .224  | .239  | .253  |
| 7.0                 | $k_x$ | .125                 | .152  | .176  | .200  | .222  | .243  | .263  | .282  | .300  | .317  | .333  |
|                     | $k_s$ | 1.046                | 1.057 | 1.068 | 1.079 | 1.091 | 1.103 | 1.115 | 1.127 | 1.138 | 1.150 | 1.161 |
|                     | $k_h$ | .050                 | .071  | .093  | .115  | .136  | .155  | .173  | .190  | .205  | .219  | .232  |
| 8.0                 | $k_x$ | .111                 | .135  | .158  | .179  | .200  | .220  | .238  | .256  | .273  | .289  | .304  |
|                     | $k_s$ | 1.040                | 1.050 | 1.060 | 1.071 | 1.081 | 1.092 | 1.103 | 1.113 | 1.124 | 1.135 | 1.145 |
|                     | $k_h$ | .044                 | .064  | .084  | .104  | .123  | .141  | .158  | .174  | .189  | .202  | .215  |
| 9.0                 | $k_x$ | .100                 | .122  | .143  | .163  | .182  | .200  | .217  | .234  | .250  | .265  | .280  |
|                     | $k_s$ | 1.036                | 1.045 | 1.054 | 1.064 | 1.073 | 1.083 | 1.093 | 1.103 | 1.113 | 1.122 | 1.132 |
|                     | $k_h$ | .040                 | .058  | .076  | .095  | .113  | .130  | .146  | .161  | .175  | .188  | .200  |
| 10.0                | $k_x$ | .091                 | .111  | .130  | .149  | .167  | .184  | .200  | .216  | .231  | .245  | .259  |
|                     | $k_s$ | 1.033                | 1.041 | 1.049 | 1.058 | 1.067 | 1.076 | 1.085 | 1.094 | 1.103 | 1.112 | 1.121 |
|                     | $k_h$ | .037                 | .053  | .070  | .087  | .104  | .120  | .135  | .149  | .163  | .175  | .187  |

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